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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	I Love You 3000
Selected Domain	Domain 1 - The Aiversecity Future-ReadyWorkforce Domain
Selected SDG Track (1–17)	SDG 4, SDG 5, SDG 8, SDG 9, SDG 10
Project / Solution Name	SkillVerse AI
Name (Leader)	GUI GUANYU
Email & Phone (Leader)	noteiinoteii2004@gmail.com & 0129307921
University Affiliation(s)	UNIVERSITI TEKNOLOGI MALAYSIA (UTM)
Number of Team Members	2 (Team Leader Included)

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence is rapidly restructuring employment faster than educational curricula can adapt. In Malaysia, the Hays Asia Salary Guide 2025 reports that while 56% of professionals already utilize AI tools at work, only 26% have received structured institutional training. Concurrently, Malaysia's Ministry of Science, Technology and Innovation (MOSTI) estimates that over 600,000 workers face severe displacement risks within the next three to five years, primarily across routine clerical administration, customer triage and basic accounting.

This technological transformation disproportionately endangers four key cohorts across the employment spectrum which include at-risk working adults facing role obsolescence, fresh graduates and career switchers navigating opaque job requirements, HR and learning-and-development managers struggling to audit enterprise skill gaps and TVET or university career advisors relying on static occupational catalogs.

Current career advisory platforms fail because they treat skill assessments as one-time, static snapshots that become obsolete the moment they are published. In an era of unpredictable AI acceleration, static frameworks widen socioeconomic divides and leave individuals ill-equipped for emerging sustainable industries. A systemic bridge is urgently requiring an adaptive platform that continuously tracks market trends, demystifies skill adjacencies and provides personalized, lifelong reskilling pathways directly aligned with the United Nations Sustainable Development Goals (SDGs 4, 5, 8, 9, and 10).

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SkillVerse AI is an adaptive workforce intelligence platform engineered to guide individuals and organizations from their current competencies to high-value future roles in the artificial intelligence era. The platform ingests candidate profiles through resume uploads, document scans or guided questionnaires, deploying an automated PII-scrubbing engine to redact demographic data and prevent gender or ethnic bias in accordance with UN SDGs 5 and 10.

Instead of relying on opaque language model predictions, SkillVerse AI executes an explainable hybrid matching formula that combines 60% macro vector semantic proximity using BAAI/bge-m3 embeddings with 40% micro itemized skill overlap. By benchmarking capabilities against European ESCO, it provides

transparent match scores and actionable skill gap analyses. To address future uncertainty, the system features dynamic macroeconomic scenarios which including moderate progress, accelerated AI disruption and green economy shifts that calibrate role viability in real time. It then synthesizes structured multiple-phase (Usually 3-Phase) reskilling roadmaps that prioritize free open-access modules alongside accredited, Malaysian HRD Corp-claimable programs, mapping every occupational trajectory to the UN SDGs.

The solution directly integrates skills acquired in SDC Phase 1 workshops. We deploy Google Gemini APIs across a circular 26-key failover pool with a five-model cascade for contextual learning mentorship and PII sanitization. For database architecture, we utilize Supabase PostgreSQL 15 leveraging the pgvector extension with HNSW indexing for rapid 1024-dimensional semantic queries. High-throughput vector embeddings are accelerated on RunPod Cloud GPU pods, supplemented by Render cloud failover to guarantee uninterrupted uptime, while live data ingestion pipelines continuously track ArXiv cs.AI research feeds and RemoteOK open job listings to calculate dynamic skill velocities (V_s). Finally, we deploy our frontend to official sponsor platform Replit for collaborative cloud prototyping, while simultaneously maintaining an independent standalone deployment on Vercel as a failover safeguard to ensure continuous platform access if Replit compute credits deplete.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

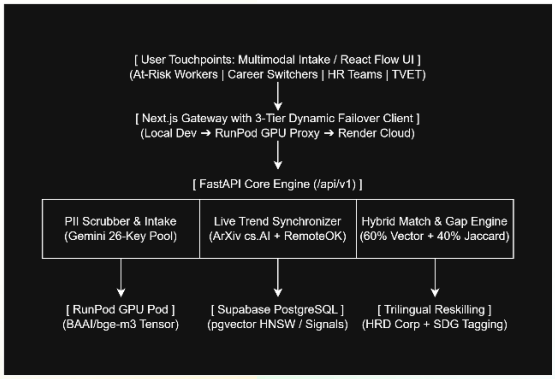
List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

SkillVerse AI is engineered as a decoupled, multi-tier cloud application connecting a high-concurrency client interface with resilient GPU vector inference and real-time labor data pipelines. The user interface is developed on Next.js 16 App Router using React Flow to render interactive and phase-based reskilling roadmaps. To guarantee uninterrupted operational access during judging, the frontend is dual deployed which is hosted on official SDC sponsor platform Replit for collaborative live prototyping and rapid reproducibility, while simultaneously maintained as an independent deployment on Vercel as a failover safeguard against credit depletion. Client interactions route through a Next.js gateway featuring a three-tier failover client coordinating between local development, RunPod GPU proxies and Render hosted web services.

The backend computational core is built with FastAPI, integrating a 26-key Google Gemini rotation pool across a five-model cascade for multimodal CV parsing, PII sanitization and contextual mentorship. Relational records, 108 standardized ESCO roles and real-time labor signals persisted in Supabase PostgreSQL 15 with pgvector HNSW indexing. The matching core executes a deterministic dual-track hybrid formula which is computing 60% Macro Vector Proximity between candidate profile paragraphs and target ESCO definitions via RunPod-hosted BAAI/bge-m3 1024-dimensional PyTorch tensors, combined with a 40% Micro Itemized Skill Overlap driven by a three-tier proximity algorithm (Exact, 60% Token Jaccard & Dense Embeddings). Real-time market trends from ArXiv cs.AI and RemoteOK continuously update skill velocities (V_s), while unindexed search queries trigger automated database expansion into trend_signals and esco_roles.

Our technical architecture heavily leverages official SDC sponsor resources. We deploy RunPod Cloud GPU pods to host BAAI/bge-m3 batch tensor inference, accelerating semantic vector distance calculations across the taxonomy. Concurrently, Replit delivers sponsor-integrated cloud hosting and collaborative development environments for building and deploying our web frontend.



3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied human-centered design-thinking methodologies by analyzing acute vulnerabilities across at-risk clerical workers, career switchers, TVET counselors and enterprise HR leaders. These empathy insights guided our architecture toward demystifying AI disruption, removing financial barriers and preventing demographic bias.

To guarantee inclusive access across diverse communities, the platform features complete trilingual localization (English, Bahasa Melayu, Simplified Chinese) with low-bandwidth, mobile-friendly interfaces. For deployment reliability, our frontend is dual deployed on sponsor platform Replit for collaborative evaluation and on Vercel as a failover safeguard against credit depletion. Operationally, our architecture replaces opaque black-box guessing with an explainable dual-track formula which combining 60% macro contextual semantic affinity with 40% micro itemized skill overlap evaluated via a three-tier proximity fast-path, streaming calculation progress via Server-Sent Events.

Crucially, our deployment preserves human agency, equipping institutional counselors with cohort deficit heatmaps and offering dynamic skill search diagnostics rather than automating hiring decisions. By coupling free open courses with Malaysian HRD Corp-claimable programs, SkillVerse AI delivers actionable, barrier-free reskilling pathways ready for Phase 2 deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Direct SDG Targets:** • **Target 4.4:** Substantially increase the number of youth and adults with relevant technical and vocational skills for employment, decent jobs and entrepreneurship. • **Target 8.5:** Achieve full and productive employment and decent work for all women and men, with equal pay for work of equal value. • **Target 8.2:** Achieve higher economic productivity through technological upgrading and innovation by steering workers into AI-augmented roles. **Indirect & Supporting SDG Targets:** • **Target 4.3:** Ensure equal access for all women and men to affordable, quality technical, vocational and tertiary education via free open-course and HRD Corp pathways. • **Target 5.1 & 5.5:** Eliminate systemic gender bias and ensure equal career progression through automated PII scrubbing. (Anonymizing gender / photos before matching.) • **Target 9.2 & 9.5:** Promote inclusive industrialization and technological innovation by building talent pipelines for sustainable tech, clean energy and responsible AI. • **Target 10.2 & 10.3:** Empower and promote universal economic inclusion while reducing inequalities of

Metric Category	Your Response
	outcome through full trilingual accessibility (EN/MS/ZH) for TVET and regional workforces.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to the Hays Asia Salary Guide 2025 and Malaysian MOSTI data, 74% of professionals lack structured AI training despite 56% daily adoption. Over 600,000 workers face imminent displacement pressure within 3 to 5 years, while institutional curriculum updates lag by 12-24 months, leaving vulnerable workers without dynamic, personalized transition pathways.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	1. **Transition Synthesis Latency:** The duration required to ingest candidate profiles, compute deterministic gaps and synthesize a verified, cost-free reskilling roadmap. (Measured in Seconds vs. Days of Manual Counseling) 2. **Real-Time Labor Signal Velocity (V_s):** Daily automated trend ingestion frequency and dynamic role self-expansion rate from live public feeds (ArXiv / RemoteOK). 3. **Algorithmic Demographic Parity:** 100% redaction rate of sensitive PII variables (gender, age, ethnicity) prior to hybrid matching.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	95% reduction in career navigation latency (generating complete, personalized pathways in under 30 seconds); 100% autonomous live trend synchronization without reliance on hard-coded occupational lists; and an enterprise cohort readiness coverage of $\geq 80\%$ across pilot TVET and corporate HR test cohorts.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [/] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

- [/] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [/] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [/] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it: -

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
BAAI/bge-m3	Pre-trained Model (Dense Semantic Embedding)	MIT License
Google Gemini APIs	Pre-trained Model / GenAI Cloud API	Google Generative AI Terms of Service
European ESCO Classification	Public Dataset (Occupational Competencies & Taxonomies)	Open Data (EU Open Data Commons Attribution 4.0)
ArXiv API (cs.AI)	Public Dataset / Live Research Data Feed	ArXiv API Terms of Use / Creative Commons
RemoteOK API	Public Dataset / Live Job Posting Feed	Public API Terms / Creative Commons
Next.js	Code Framework (Web Frontend & Gateway)	MIT License
FastAPI	Code Framework (Python Backend Engine)	MIT License

Supabase	Code / PostgreSQL Database Extension	PostgreSQL Open-Source License
Replit Platform	Cloud Hosting & Collaborative Dev Platform (SDC Sponsor)	Terms of Service (Replit)
Vercel Platform	Cloud Hosting (Edge Web Deployment Failover)	Terms of Service (Vercel)
RunPod Cloud Platform	Cloud Compute / Primary Backend Service & GPU Acceleration (SDC Sponsor)	Terms of Service (RunPod)
Render Cloud Platform	Cloud Hosting / Backup Secondary Backend Web Service (Identical Failover Node)	Terms of Service (Render)

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns: -

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



GUI GUANYU

23/10/2026